

Ball and Beam Experiment: Prelab Questions

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September 24, 2026 – Week 9, Thursday 10 am

Background

Time-Domain Specifications

Specified steady-state error, 0.04% settling time, and percentage overshoot:

$$\begin{aligned} |e_{ss}| &\leq 0.005\text{m} \\ t_s &= 3.5\text{s} \\ PO &= 10.0\% \end{aligned}$$

Key Equations

Equation of motion and mass moment of inertia for the ball (Equation 2.18):

$$\frac{d^2}{dt^2}x(t) = \frac{m_b g r_{arm} r_b^2}{L_{beam} (m_b r_b^2 + J_b)} \theta_l(t) \quad J_b = \frac{2m_b r_b^2}{5}$$

Standard second-order system transfer function (Equation 2.39):

$$\frac{X(s)}{X_d(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n + \omega_n^2}$$

Peak time (Equation 2.30) and percent overshoot (Equation 2.31):

$$t_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}} \quad PO = 100 \exp\left(\frac{-\pi\zeta}{\sqrt{1 - \zeta^2}}\right)$$

Ideal PD control design parameters (Equation 2.40) and (Equation 2.41):

$$z = \frac{\omega_n^2}{K_{bb}K_c} \quad K_c = \frac{2\zeta\omega_n}{K_{bb}}$$

Practical PD control, $\omega_f = 6.28 \text{ rad/s}$. Characteristic equation for closed-loop (Equation 2.52):

$$s^3 + s^2\omega_f + (K_{bb}K_c\omega_f + K_{bb}K_c z)s + K_{bb}K_c z\omega_f$$

Location of compensator zero, pole and the pole time constant (Equations 2.48, 2.49, 2.54):

$$z_c = \frac{-z\omega_f}{z + \omega_f} \quad p_c = -\omega_f \quad T_p = \frac{1}{\omega_f - 2\zeta\omega_n}$$

BB01 System Specifications

Symbol	Description	MATLAB Variable	Value
L_{beam}	Beam length	<code>L_beam</code>	0.4255 m
r_{arm}	Gear shaft to joint	<code>r_arm</code>	0.0254 m
r_b	Radius of ball	<code>r_ball</code>	0.0127 m
m_b	Mass of ball	<code>m_ball</code>	0.064 kg

Question 1

The mass moment of inertia:

$$J_b = \frac{2(0.064)(0.0127^2)}{5} = 4.1290 \times 10^{-6} \text{ kg} \cdot \text{m}^2$$

We substitute values into Equation 2.18 to get the constant term K_{bb} :

$$K_{bb} = \frac{(0.064)(9.81)(0.0254)(0.0127^2)}{0.4255(0.064(0.0127)^2 + 4.129 \cdot 10^{-6})} = 0.4183 \text{ m/s}^2/\text{rad}$$

Question 5

Rearranging Equation 2.31 and using $PO = 10.0\%$, the required damping ratio is:

$$\ln^2 \left(\frac{PO}{100} \right) = \frac{\pi^2 \zeta^2}{1 - \zeta^2} \quad \therefore \zeta = \sqrt{\frac{\ln^2 \left(\frac{PO}{100} \right)}{\pi^2 + \ln^2 \left(\frac{PO}{100} \right)}} = 0.5912$$

Then rearrange Equation 2.30 and substitute in $t_p = 3.5$:

$$t_p^2 = \frac{\pi^2}{\omega_n^2 (1 - \zeta^2)} \quad \therefore \omega_n = \sqrt{\frac{\pi^2}{t_p^2 (1 - \zeta^2)}} = 1.1129 \text{ rad/s}$$

Question 11

The required compensator gain using Equation 2.41:

$$K_c = \frac{2(0.5912)(1.1129)}{0.4183} = 3.1456 \text{ s/m}$$

The location of the zero using Equation 2.40:

$$z = \frac{1.1129^2}{0.4183(3.1456)} = 0.9413 \text{ s}^{-1}$$

Question 12

For the practical PD controller we set a high-pass corner frequency $\omega_f = 2\pi \text{ rad/s}$. Comparing the s^0 and s^1 terms in Equation 2.52 and Equation 2.31:

$$K_{bb}K_c z = \omega_n^2 \quad K_{bb}K_c \left(1 + \frac{z}{\omega_f} \right) = 2\zeta\omega_n$$

We rearrange terms to find the formula for K_c :

$$K_c = \frac{2\zeta\omega_n - \omega_n^2/2\pi}{K_{bb}} = \frac{2(0.5912)(1.1129) - (1.1129^2)/2\pi}{0.4183} = 2.6743 \text{ s/m}$$

Then we can get:

$$z = \frac{\omega_n^2}{K_{bb}K_c} = \frac{1.1129^2}{0.4183(2.6743)} = 1.1071 \text{ s}^{-1}$$

The location of the zero is:

$$z_c = \frac{-z\omega_f}{z + \omega_f}$$

We use Equation 2.54 to determine the pole time constant:

$$T_p = \frac{1}{\omega_f - 2\zeta\omega_n} = \frac{1}{2\pi - 2(0.5192)(1.1129)} = 0.2013 \text{ s}$$